

CSC 4402: Assignment 1

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1.

a) Primary keys

branch: branch_name

customer: ID

loan: loan_number

borrower: ID, loan_number

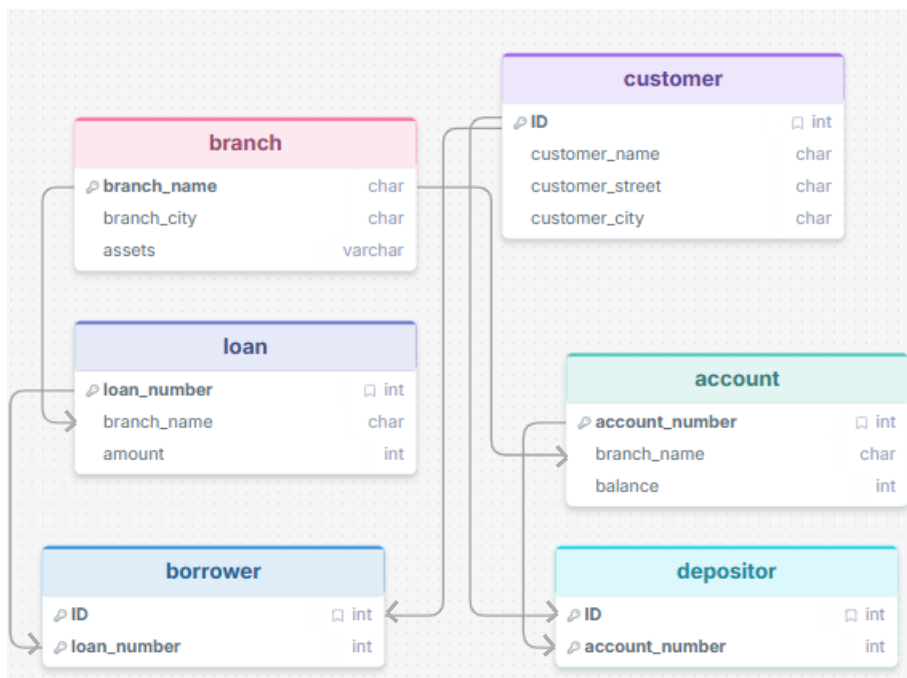
account: account_number

depositor: ID, account_number

b)

$\Pi_{employee.name} (\sigma_{employee.city = "London" \wedge works.salary < 70000} (employee \bowtie_{employee.ID = works.ID} works))$

2.



3.

a)

update instructor i

set i.salary = i.salary * 1.08

where i.salary in (select min(i.salary)

from i

where i.dept_name = 'Comp. Sci.');

b)

```
select dept_name, avg (salary)
from instructor
where avg (salary) between 52000 and 75000
group by dept_name;
```

4.

a)

```
Select ID
From works
Where salary > all      (select salary
                        from works
                        where company_name = 'Bank of America');
```

b)

The two separate updates are not atomic, as they are executed sequentially. It is possible that one query would be executed, but not the other, in a fail/crash, leading to potential double salary bumps.

Correct query:

Update works

Set salary = salary * 1.09 - salary * 0.06 * (salary * 1.09 > 100000)

Where ID in (Select manager_id from manages)

and company_name = 'JP Morgan Chase';